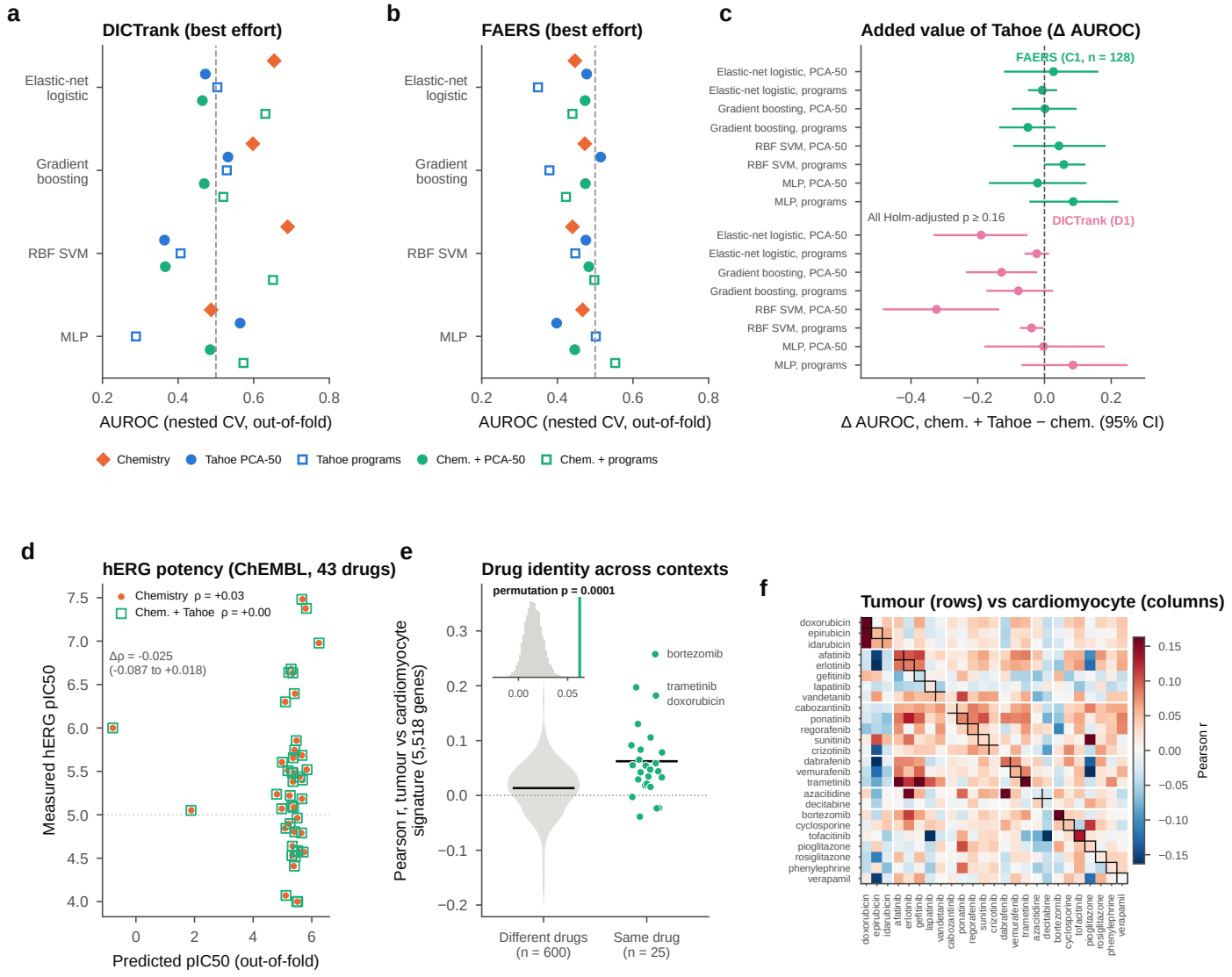
a
DICTrank (best effort)
Elastic-net logistic
Gradient boosting
RBF SVM
MLP
0.2
0.4
0.6
0.8
AUROC (nested CV, out-of-fold)
b
FAERS (best effort)
Elastic-net logistic
Gradient boosting
RBF SVM
MLP
0.2
0.4
0.6
0.8
AUROC (nested CV, out-of-fold)
c
Added value of Tahoe (Δ AUROC)
FAERS (C1, n = 128)
Elastic-net logistic, PCA-50
Elastic-net logistic, programs
Gradient boosting, PCA-50
Gradient boosting, programs
RBF SVM, PCA-50
RBF SVM, programs
MLP, PCA-50
MLP, programs
All Holm-adjusted p ≥ 0.16
DICTrank (D1)
Elastic-net logistic, PCA-50
Elastic-net logistic, programs
Gradient boosting, PCA-50
Gradient boosting, programs
RBF SVM, PCA-50
RBF SVM, programs
MLP, PCA-50
MLP, programs
−0.4
−0.2
0.0
0.2
Δ AUROC, chem. + Tahoe − chem. (95% CI)
Chemistry
Tahoe PCA-50
Tahoe programs
Chem. + PCA-50
Chem. + programs
d
hERG potency (ChEMBL, 43 drugs)
Chemistry ρ = +0.03
Chem. + Tahoe ρ = +0.00
Δρ = -0.025
(-0.087 to +0.018)
Measured hERG pIC50
Predicted pIC50 (out-of-fold)
e
Drug identity across contexts
permutation p = 0.0001
0.00
0.05
bortezomib
trametinib
doxorubicin
Pearson r, tumour vs cardiomyocyte signature (5,518 genes)
Different drugs (n = 600)
Same drug (n = 25)
f
Tumour (rows) vs cardiomyocyte (columns)
doxorubicin
epirubicin
idarubicin
afatinib
erlotinib
gefitinib
lapatinib
vandetanib
cabozantinib
ponatinib
regorafenib
sunitinib
crizotinib
dabrafenib
vemurafenib
trametinib
azacitidine
decitabine
bortezomib
cyclosporine
tofacitinib
pioglitazone
rosiglitazone
phenylephrine
verapamil
Pearson r